

Foundations of Linear discriminant analysis

[Linear discriminant analysis]

1.0 Content Block

Multivariate normality: Independent variables are normal for each level of the grouping variable. Homogeneity of variance/covariance (homoscedasticity): Variances among group variables are the same across levels of predictors. Can be tested with Box's M statistic. It has been suggested, however, that linear discriminant analysis be used when covariances are equal, and that quadratic discriminant analysis may be used when covariances are not equal. Independence: Participants are assumed to be randomly sampled, and a participant's score on one variable is assumed to be independent of scores on that variable for all other participants. It has been suggested that discriminant analysis is relatively robust to slight violations of these assumptions, and it has also been shown that discriminant analysis may still be reliable when using dichotomous variables (where multivariate normality is often violated).

2.0 Content Block

$$\frac{1}{2} (x - \mu_0)^T \Sigma_0^{-1} (x - \mu_0) + \frac{1}{2} \ln |\Sigma_0| - \frac{1}{2} (x - \mu_1)^T \Sigma_1^{-1} (x - \mu_1) - \frac{1}{2} \ln |\Sigma_1| > T \quad \left\{ \text{displaystyle } \frac{1}{2} (\vec{x} - \vec{\mu}_0)^T \Sigma_0^{-1} (\vec{x} - \vec{\mu}_0) + \frac{1}{2} \ln |\Sigma_0| - \frac{1}{2} (\vec{x} - \vec{\mu}_1)^T \Sigma_1^{-1} (\vec{x} - \vec{\mu}_1) - \frac{1}{2} \ln |\Sigma_1| > T \right\}$$

3.0 Content Block

Comparison to logistic regression Discriminant function analysis is very similar to logistic regression, and both can be used to answer the same research questions. Logistic regression does not have as many assumptions and restrictions as discriminant analysis. However, when discriminant analysis' assumptions are met, it is more powerful than logistic regression. Unlike logistic regression, discriminant analysis can be used with small sample sizes. It has been shown that when sample sizes are equal, and homogeneity of variance/covariance holds, discriminant analysis is more accurate. Despite all these advantages, logistic regression has none-the-less become the common choice, since the assumptions of discriminant analysis are rarely met.

4.0 Content Block

$x \rightarrow T \Sigma_i^{-1} \mu_i \rightarrow i = \mu_i \rightarrow i T \Sigma_i^{-1} x \rightarrow \left\{ \text{displaystyle } \vec{x}^T \Sigma_i^{-1} \vec{\mu}_i = \vec{\mu}_i^T \Sigma_i^{-1} \vec{x} \right\}$ because both sides are scalar and transpose to each other (Σ_i is Hermitian) and the above decision criterion becomes a threshold on the dot product

5.0 Content Block

Linear discriminant analysis (LDA), normal discriminant analysis (NDA), canonical variates analysis (CVA), or discriminant function analysis is a generalization of Fisher's linear discriminant, a method used in statistics and other fields, to find a linear combination of features that characterizes or separates two or more classes of objects or events. The resulting combination may be used as a linear classifier, or, more commonly, for dimensionality reduction before later classification. LDA is closely related to analysis of variance (ANOVA) and regression analysis, which also attempt to express one dependent variable as a linear combination of other features or measurements. However, ANOVA uses categorical independent variables and a continuous dependent variable, whereas discriminant analysis has continuous independent variables and a categorical dependent variable (i.e. the class label). Logistic regression and probit regression are more similar to LDA than ANOVA is, as they also explain a categorical variable by

the values of continuous independent variables. These other methods are preferable in applications where it is not reasonable to assume that the independent variables have a normal distribution, which is a fundamental assumption of the LDA method. LDA is also closely related to principal component analysis (PCA) and factor analysis in that they both look for linear combinations of variables which best explain the data. LDA explicitly attempts to model the difference between the classes of data. PCA, in contrast, does not take into account any difference in class, and factor analysis builds the feature combinations based on similarities rather than differences. Discriminant analysis is also different from factor analysis in that it is not an interdependence technique: a distinction between independent variables and dependent variables (also called criterion variables) must be made. LDA works when the measurements made on independent variables for each observation are continuous quantities. When dealing with categorical independent variables, the equivalent technique is discriminant correspondence analysis. Discriminant analysis is used when groups are known a priori (unlike in cluster analysis). Each case must have a score on one or more quantitative predictor measures, and a score on a group measure. In simple terms, discriminant function analysis is classification - the act of distributing things into groups, classes or categories of the same type.

6.0 Content Block

$$\mathbf{x} \rightarrow \mathbf{T} \Sigma \mathbf{0} - \mathbf{1} \mathbf{x} \rightarrow = \mathbf{x} \rightarrow \mathbf{T} \Sigma \mathbf{1} - \mathbf{1} \mathbf{x} \rightarrow \{\displaystyle \{\vec{x}\}^{\mathrm{T}}\} \Sigma_{0}^{-1} \{\vec{x}\} = \{\vec{x}\}^{\mathrm{T}} \} \Sigma_{1}^{-1} \{\vec{x}\}$$

7.0 Content Block

History The original dichotomous discriminant analysis was developed by Sir Ronald Fisher in 1936. It is different from an ANOVA or MANOVA, which is used to predict one (ANOVA) or multiple (MANOVA) continuous dependent variables by one or more independent categorical variables. Discriminant function analysis is useful in determining whether a set of variables is effective in predicting category membership.

8.0 Content Block

Earth science This method can be used to separate the alteration zones. For example, when different data from various zones are available, discriminant analysis can find the pattern within the data and classify it effectively.

9.0 Content Block

Biomedical studies The main application of discriminant analysis in medicine is the assessment of severity state of a patient and prognosis of disease outcome. For example, during retrospective analysis, patients are divided into groups according to severity of disease – mild, moderate, and severe form. Then results of clinical and laboratory analyses are studied to reveal statistically different variables in these groups. Using these variables, discriminant functions are built to classify disease severity in future patients. Additionally, Linear Discriminant Analysis (LDA) can help select more discriminative samples for data augmentation, improving classification performance. In biology, similar principles are used in order to classify and define groups of different biological objects, for example, to define phage types of *Salmonella enteritidis* based on Fourier transform infrared spectra, to detect animal source of *Escherichia coli* studying its virulence factors etc.